



OPEN ACCESS

EDITED BY

Lucia Maddaloni,
Sapienza University of Rome, Italy

REVIEWED BY

Charilaos Dimosthenopoulos,
Laiko General Hospital of Athens, Greece
Saniye Bilici,
Gazi University Faculty of Health
Sciences, Türkiye

*CORRESPONDENCE

Rafael Fernández-Demeneghi
✉ rafael_demeneghi@hotmail.com

†These authors have contributed equally
to this work and share first authorship

RECEIVED 19 January 2026
REVISED 19 February 2026
ACCEPTED 24 February 2026
PUBLISHED 13 March 2026

CITATION

Fernández-Demeneghi R,
Sánchez-Jiménez MA,
Huerta-Canseco C, Vargas-Moreno I,
Rosas-Sánchez GU,
Ramírez-Rodríguez R, Páez-Huerta G
and Álvarez-Ramírez MM (2026) The
Milpa diet: a functional, sustainable
pattern for human and planetary health.
Front. Nutr. 13:1790861.
doi: 10.3389/fnut.2026.1790861

COPYRIGHT

© 2026 Fernández-Demeneghi,
Sánchez-Jiménez, Huerta-Canseco,
Vargas-Moreno, Rosas-Sánchez,
Ramírez-Rodríguez, Páez-Huerta and
Álvarez-Ramírez. This is an open-access
article distributed under the terms of the
[Creative Commons Attribution License
\(CC BY\)](https://creativecommons.org/licenses/by/4.0/). The use, distribution or
reproduction in other forums is
permitted, provided the original author(s)
and the copyright owner(s) are credited
and that the original publication in this
journal is cited, in accordance with
accepted academic practice. No use,
distribution or reproduction is permitted
which does not comply with these terms.

The Milpa diet: a functional, sustainable pattern for human and planetary health

Rafael Fernández-Demeneghi^{1,2*†},
Martha Alicia Sánchez-Jiménez^{3†}, César Huerta-Canseco⁴,
Isidro Vargas-Moreno^{1,2}, Gilberto Uriel Rosas-Sánchez^{5,6},
Rodrigo Ramírez-Rodríguez⁷, Gabriela Páez-Huerta² and
María Magdalena Álvarez-Ramírez²

¹Instituto de Investigaciones en Comportamiento Alimentario y Nutrición, Universidad de Guadalajara, Ciudad Guzmán, Mexico, ²Facultad de Nutrición, Campus Xalapa Universidad Veracruzana, Xalapa, Mexico, ³Programa de doctorado en Psicología con Orientación en Calidad de Vida y Salud, Universidad de Guadalajara, Ciudad Guzmán, Mexico, ⁴Licenciatura en Nutrición, Vicerrectoría de Ciencias de la Salud, Universidad de Monterrey, San Pedro Garza García, Mexico, ⁵Departamento de Ciencias de la Tierra y de la Vida, Centro Universitario de Los Lagos, Universidad de Guadalajara, Lagos de Moreno, Mexico, ⁶Programa de Estancias Posdoctorales por México, SECIHTI, Centro Universitario de Los Lagos, Universidad de Guadalajara, Lagos de Moreno, Mexico, ⁷Instituto Politécnico Nacional, Ciudad de México, Mexico

KEYWORDS

agroecology, food sovereignty, indigenous knowledge, Milpa diet, planetary health, precision nutrition

Introduction: the Milpa diet concept and definition

The synchronization of metabolic dysfunction and planetary degradation constitutes the defining challenge of the Anthropocene. We face a global syndemic—comprising obesity, undernutrition, and climate change—where unhealthy dietary transitions and unsustainable food systems act not merely as consequences, but as a mutually reinforcing feedback loop (1–3). This triple crisis creates a synergistic burden that undermines the viability of sustainable food systems. As environmental and public health degrade, it triggers a regressive cycle where individuals are increasingly subjected to unhealthy food environments, further exacerbating the initial drivers of the syndemic. The magnitude of this crisis is evident: currently, over 2 billion adults present with overweight or obesity (4), while a staggering proportion of mortality attributable to cardiovascular diseases (30%), type 2 diabetes (23%), and chronic kidney disease (20.7%) is directly underpinned by dietary risks (1, 5, 6).

This public health emergency is inextricably linked to an agri-food model dependent on high-yield monocultures. While this industrial paradigm has maximized caloric availability, it has compromised nutritional diversity and ecosystem integrity (7, 8). The expansion of simplified agricultural systems now drives 80% of deforestation and consumes 70% of freshwater (9), generating a “hidden hunger” where caloric abundance coexists with micronutrient deficiencies (10). Consequently, nutrition cannot be approached through a reductionist lens of isolated nutrients; it must be understood as a multidimensional phenomenon shaped by environmental, social, and economic determinants (11, 12).

In this context, the Milpa Diet—rooted in Mesoamerican agricultural traditions—transcends mere cultural grounding to become a robust biocultural system and a complex nutritional ecosystem. Far from being a relic of the past, the Milpa represents a sophisticated, nutrient-dense, and bioactive-rich pattern that aligns with the modern

principles of integrative and precision nutrition (13, 14). Historically, dietary interventions have been framed through a reactive, pathogenic lens focused strictly on risk-factor reduction and disease prevention. The Milpa bridges this theoretical gap. Based on the polyculture of maize, beans, squash, and chili, this system certainly supports metabolic health and prevents non-communicable diseases (NCDs) through synergistic biochemical mechanisms rather than isolated compounds (15–17); however, its theoretical grounding goes much further.

Moving beyond the concept of a mere “meal plan,” the Milpa paradigm embodies a true salutogenic model—focused proactively on the origins of health and the holistic development of wellbeing rather than merely reacting to disease. Its structure fosters food sovereignty, biodiversity preservation, and climate adaptation through efficient resource cycling and agroecological practices (18). This opinion article explores the relevance of the Milpa system not only as a cultural heritage but as a scalable, evidence-based blueprint for sustainable diets (Figure 1). We posit that integrating this indigenous knowledge system offers a strategic, viable opportunity to address the dual burden of chronic disease and environmental degradation, advancing the transition toward resilient, equitable, and sustainable food systems.

The Milpa system: concept, definition, and nutritional structure

The term Milpa derives from the Nahuatl words *milli* (“sown plot”) and *pan* (“on top of”), designating an ancestral Mesoamerican agricultural system that originated between 4,500- and 3,500-years BP (18, 19). However, functionally, it represents more than a cultivation technique; it is a dynamic biocultural complex adapted to diverse edaphoclimatic conditions. Its nutritional core consists of the “Mesoamerican Triad”: maize (*Zea mays* L.), common bean (*Phaseolus vulgaris* L.), and squash (*Cucurbita pepo* L.), which form a symbiotic structural and nutritional base (20). Depending on regional ecosystems, this polyculture integrates a diverse array of endemic species, functioning as an *in-situ* germplasm bank. These include chili peppers (*Capsicum annuum* L.), tomatoes (*Lycopersicon esculentum*), tomatillo (*Physalis* spp.), and chilacayote (*Cucurbita ficifolia*). The system also incorporates amaranth (*Amaranthus* spp.), nopal cactus (*Opuntia ficus-indica*), and nutrient-dense wild greens known as quelites—such as chipilín (*Crotalaria longirostrata*), *Amaranthus hybridus*, *Portulaca oleracea*, and *Dysphania ambrosioides*. This biodiversity is complemented by tubers like sweet potato (*Ipomoea batatas*) and cassava (*Manihot esculenta*) (15, 18–21).

1 Agroecological synergy and bio-efficiency

From an agroecological perspective, the Milpa thrives as a mixed polyculture within a space of synergistic interactions, optimizing the capture of solar radiation, water, and soil nutrients. The vertical architecture of maize provides support for climbing

beans, while the broad leaves of squash cover the soil, retaining moisture and suppressing competitive weeds (15, 19). This structure optimizes the “Land Equivalent Ratio” (LER), resulting in a food system rich in proteins, fiber, vitamins, and bioactive compounds, produced with high resource efficiency (17).

2 The technological component: nixtamalization

Crucially, the nutritional architecture of the Milpa diet cannot be understood solely from its raw agricultural products; it relies fundamentally on ancestral food-processing technologies, specifically nixtamalization. This alkaline thermal treatment of maize (cooking with calcium hydroxide) breaks down the hemicellulose of the pericarp, significantly enhancing the bioavailability of niacin (vitamin B3) and calcium, while modifying the protein matrix to improve digestibility (22). Thus, the Milpa system fuses agricultural biodiversity with culinary biotechnology to deliver a complete protein profile and a micronutrient-dense diet.

3 The Milpa diet: a functional nutritional matrix

Derived from this agricultural model, the Milpa Diet is proposed not merely as a set of ingredients but as a healthy, sustainable, and culturally relevant dietary pattern. Its nutritional axis relies on the synergistic consumption of maize, beans, squash, and chili (13, 16). These components create a complementary nutritional matrix characterized by a high density of micronutrients, diverse fiber types, and a complex profile of bioactive compounds:

- **Maize (*Zea mays*):** providing an energy foundation, it provides complex carbohydrates and resistant starch. Enriched by nixtamalization, it becomes a key source of bioavailable calcium. Furthermore, pigmented varieties (blue/red) function as potent reservoirs of anthocyanins—particularly cyanidin-3-glucoside—and phenolic acids (ferulic acid), which are linked to antioxidant and anti-inflammatory pathways (17, 22).
- **Common beans (*Phaseolus vulgaris*):** these legumes are critical for the protein architecture of the diet. They act as the biochemical complement to maize. In contrast, maize is low in lysine and tryptophan; beans are rich in these essential amino acids, providing protein quality comparable to animal sources. Additionally, they are rich in iron, zinc, folate, and non-digestible fermentable carbohydrates (prebiotics), as well as flavonoids such as quercetin and kaempferol (23, 24).
- **Squash (*Cucurbita* spp.):** this crop offers a dual nutritional contribution. The seeds (pepitas) provide a dense source of plant-based proteins, zinc, magnesium, and healthy lipids (mono- and polyunsaturated fatty acids), including phytosterols. Conversely, the pulp serves as a primary source of carotenoids (β -carotene, lutein, zeaxanthin), vital for ocular and immune health (25, 26).

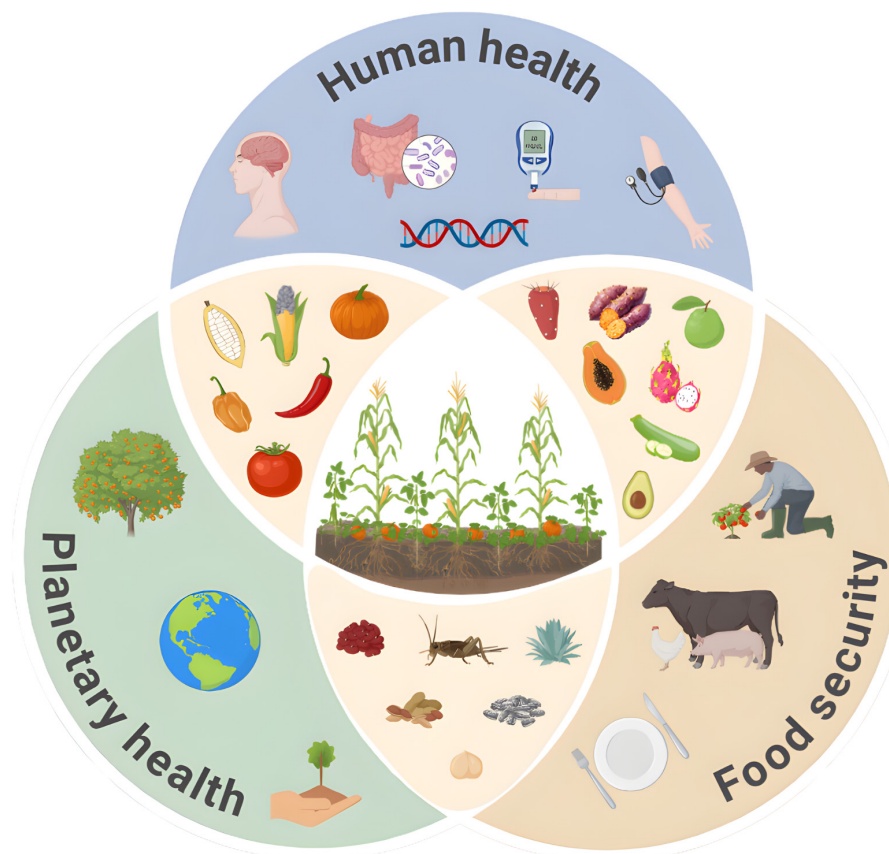


FIGURE 1

The Milpa paradigm as a synergistic framework for human health, planetary health, and food security. The central intersection represents the core Mesoamerican triad (maize, beans, and squash) cultivated as an agroecological polyculture. The overlapping spheres illustrate the multidimensional impacts of this biocultural system. Human health is supported by diverse, nutrient-dense foods that positively modulate the gut microbiome, metabolic markers, and genomic expression. Planetary health is promoted through biodiversity conservation, sustainable land use, and climate adaptation. Food security is achieved via resilient local agriculture, culturally relevant protein sources (including backyard livestock and edible insects), and food sovereignty. Together, these elements form a holistic, salutogenic nutritional model.

- Chili peppers (*Capsicum* spp.): far from being a mere condiment, chili peppers act as metabolic modulators. They are exceptionally rich in Vitamins A, C, and E. Their pungency derives from capsaicinoids, compounds with recognized thermogenic and metabolic-regulating properties, which work in tandem with their high flavonoid content (27).

This combination generates a distinctive nutritional phenotype that modulates metabolic health via multiple bioactive pathways. To illustrate the specific functional contribution of this biodiversity, Table 1 provides a detailed summary of the bioactive compounds and physiological effects associated with the key Mesoamerican species constituting the Milpa Diet.

4 Dietary structure and composition

This core triad is dynamically complemented by eight peripheral food groups: vegetables, fruits, whole grains, oilseeds, unsaturated fats, tubers, natural sweeteners, and animal protein in moderation. The regimen prioritizes water consumption while strictly limiting ultra-processed products (16, 17). Collectively, the Milpa Diet constitutes a predominantly plant-based yet fundamentally omnivorous framework. It is defined by a sophisticated macronutrient architecture—rich in plant proteins, complex carbohydrates, and healthy fats—and reinforced by a high fiber content and a diverse phytochemical spectrum.

The Milpa diet and health: mechanisms and bioactive architecture

Contemporary nutritional science effectively validates the therapeutic potential of traditional dietary patterns rooted in whole, minimally processed foods. The Milpa diet aligns rigorously with this paradigm, offering a functional food matrix characterized by a low glycemic load, high fiber density, and a diverse spectrum of bioactive compounds. Unlike reductionist approaches, the therapeutic potency of this diet relies on the biochemical synergy of its components—a “natural polypharmacy” effect that targets metabolic dysregulation through multiple signaling pathways (17, 23).

TABLE 1 Mesoamerican foods included in the Milpa Diet, their bioactive compounds, and health effects.

Food group/common name (scientific name)	Bioactive compounds	Health effect	Reference
Maize (<i>Zea mays</i> L.)	Polyphenols (ferulic acid), anthocyanins, carotenoids, dietary fiber, phytosterols, linoleic acid (ω -6), oleic acid (ω -9).	Antioxidant, hypocholesterolemic, glycemic control.	(17)
Pumpkin (<i>Cucurbita pepo</i> L.)	Phytosterols, cucurbitacins, tocopherols, linoleic acid (ω -6), oleic acid (ω -9), palmitic acid.	Hypocholesterolemic, antioxidant.	(17)
Common bean (<i>Phaseolus vulgaris</i> L.)	Polyphenols, flavonoids, saponins, lectins, resistant starch, linoleic acid, α -linolenic acid (ω -3).	Glycemic control, hypocholesterolemic, intestinal health.	(17)
Vegetables and greens			
Epazote (<i>Chenopodium ambrosioides</i> L.)	Flavonoids, terpenes, polyphenols, fatty acids.	Antiparasitic, antioxidant, anti-inflammatory, immunomodulatory, glycemic control, anticancer	(57)
Habanero pepper (<i>Capsicum chinense</i> Jacq.)	Capsaicinoids, carotenoids, polyphenols.	Anti-inflammatory, antioxidant.	(58)
Huanzontle (<i>Chenopodium nuttalliae</i>)	Polyphenols, flavonoids, carotenoids.	Anti-inflammatory, antioxidant, hypocholesterolemic.	(59)
Nopal (<i>Opuntia ficus-indica</i> L.)	Flavonoids, mucilages, soluble fiber, linoleic acid (ω -6), α -linolenic acid (ω -3).	Anti-inflammatory, antioxidant, glycemic control, antimicrobial, neuroprotective.	(60)
Huitlacoche (<i>Ustilago maydis</i>)	β -glucans, polyphenols, linoleic acid (ω -6), oleic acid (ω -9).	Immunomodulatory, antioxidant.	(61)
Tomato (<i>Solanum lycopersicum</i> L.)	β -carotene, lycopene, tocopherol, phenolic acids, flavonoids, anthocyanin.	Antioxidant, hypocholesterolemic, antiplatelet aggregation, antithrombotic, antihypertensive, hypoglycemic, anticancer.	(62)
Tomatillo (<i>Physalis philadelphica</i> Lam.)	Flavonoids, polyphenols, carotenoids, phytosterols.	Antitumoral, anti-inflammatory.	(63)
Zucchini flower (<i>Cucurbita pepo</i> L.)	Flavonoids, carotenoids, polyphenols.	Antioxidant.	(64)
Chayote (<i>Sechium edule</i>)	Carotenoids, polyphenols, flavonoids.	Antioxidant, hypocholesterolemic.	(65)
Pepper (<i>Capsicum annum</i> L.)	Capsaicinoids, carotenoids, polyphenols, flavonoids.	Anticancer, anti-inflammatory, antidiabetic, antihypertensive.	(66)
Romeritos (<i>Suaeda torreyana</i>)	Polyphenols, carotenoids, fiber, linoleic acid (ω -6).	Antioxidant, hypocholesterolemic.	(67)
Purslane (<i>Portulaca oleracea</i>)	Flavonoids, alkaloids, terpenoids, α -linolenic acid (ω -3).	Anti-inflammatory, antioxidant, immunomodulatory.	(68)
Fruits			
Black sapote (<i>Diospyros digyna</i> Jacq.)	Polyphenols, proanthocyanidins, flavonoids.	Antioxidant.	(69)
Cactus Berry (<i>Opuntia humifusa</i>)	Betalains, polyphenols, anthocyanins.	Antioxidant, hypocholesterolemic.	(70)
Canistel (<i>Pouteria campechiana</i>)	Carotenoids, polyphenols, flavonoids.	Antioxidant, anti-inflammatory, hypolipidemic, anticancer, glycemic control.	(71)
Cherimoya (<i>Annona cherimola</i> Mill.)	Polyphenols, flavanols, procyanidins.	Antioxidant, anticancer, antiallergy.	(72)
Dragon fruit (<i>Hylocereus undatus</i>)	Betalains, polyphenols, fiber, anthocyanins, Linoleic acid (ω -6), α -linolenic acid (ω -3).	Antioxidant, hypocholesterolemic, glycemic control, cardioprotective, anti-inflammatory.	(73)
Mamey (<i>Pouteria sapota</i>)	Carotenoids, polyphenols.	Antioxidant, cardioprotective, anti-inflammatory, anticancer.	(74)

(Continued)

TABLE 1 (Continued)

Food group/common name (scientific name)	Bioactive compounds	Health effect	Reference
Mexican hawthorn (<i>Crataegus mexicana</i>)	Polyphenols, proanthocyanidins, vitamin C, flavonoids, triterpenic glycosides.	Antioxidant, cardioprotective.	(75)
Papaya (<i>Carica papaya</i> L.)	Carotenoids, papain, polyphenols, saponins.	Antioxidant, digestive, anti-inflammatory, antimicrobial, anticancer, glycemic control.	(76)
Pitaya (<i>Stenocereus stellatus</i>)	Betalains, polyphenols, flavonoids, fiber.	Antioxidant, hypocholesterolemic, intestinal health, antihypertensive, anti-inflammatory.	(77)
Red mombin (<i>Spondias purpurea</i> L.)	Polyphenols, carotenoids, saponins, polysaccharides.	Antioxidant, anti-inflammatory, glycemic control, hypocholesterolemic, ulcer protective, hepatoprotective, antiarthritic, antihypertensive, antiepileptic.	(78)
Sapodilla (<i>Manilkara zapota</i>)	Carotenoids, polyphenols.	Antioxidant, anticancer, antimicrobial, anti-inflammatory, antispasmodic, anti-aging, hepatoprotective.	(79)
Xoconostle (<i>Opuntia joconostle</i>)	Betalains, flavonoids, polyphenols, fiber, linoleic acid (ω -6), oleic acid (ω -9), tocopherols.	Glycemic control, hypocholesterolemic, antioxidant, anti-inflammatory.	(80)
Soursop (<i>Annona muricata</i> L.)	Acetogenins, flavonoids, alkaloids, polyphenols.	Antioxidant, antimicrobial, antinociceptive, antihypertensive, glycemic control, anticancer.	(81)
Yellow mombin (<i>Spondias mombin</i> L.)	Polyphenols, flavonoids.	Antioxidant, anti-inflammatory.	(82)
Whole grains			
Amaranth (<i>Amaranthus</i> spp.)	Squalene, bioactive peptides, polyphenols, oleic acid, linoleic acid, anthocyanins, betalain, carotenoids, flavonoids.	Hypocholesterolemic, gastroprotective, anti-inflammatory, anticancer.	(83)
Oilseeds and legumes			
Chia (<i>Salvia hispanica</i> L.)	α -Linolenic acid (ω -3), linoleic acid (ω -6), soluble fiber, polyphenols, flavonoids, carotenoids.	Antioxidant, hypotriglyceridemic, hypocholesterolemic, antihypertensive, anticancer, glycemic control.	(84)
Sunflower seed (<i>Helianthus annuus</i>)	Polyphenols, phytosterols, flavonoids, carotenoids, linoleic acid (ω -6), oleic acid (ω -9).	Antioxidant, anti-inflammatory, glycemic control, antimicrobial, antihypertensive.	(85)
Pumpkin seed (<i>Cucurbita pepo</i>)	Phytosterols, cucurbitacins, tocotrienols, oleic acid (ω -9).	Hypocholesterolemic, antioxidant.	(17)
Cocoa (<i>Theobroma cacao</i>)	Flavonoids, epicatechin, proanthocyanidins, saponins, oleic acid (ω -9).	Antioxidant, glycemic control, antihypertensive, anticancer, hypocholesterolemic, neuroprotector.	(86)
Peanut (<i>Arachis hypogaea</i>)	Dietary fiber, phytosterols, polyphenols, oleic acid (ω -9), linoleic acid (ω -6), resveratrol, flavonoids.	Hypocholesterolemic, glycemic control, intestinal health, antihypertensive, anticancer, neuroprotector.	(87)
Haba (<i>Vicia faba</i> L.)	Polyphenols, flavonoids, fiber, linoleic acid (ω -6), carotenoids.	Glycemic control, antioxidant, intestinal health, antihypertensive, anticancer.	(88)
Lentil (<i>Lens culinaris</i> L.)	Polyphenols, flavonoids, tannins, bioactive peptides, linoleic acid (ω -6).	Glycemic control, antioxidant, hypocholesterolemic, anticancer.	(89)
Unsaturated oils			
Avocado (<i>Persea americana</i> Mill.)	Oleic acid, linoleic acid, polyphenol, phytosterols, carotenoids, tocopherols.	Hypocholesterolemic, antioxidant, cardioprotective.	(90)

(Continued)

TABLE 1 (Continued)

Food group/common name (scientific name)	Bioactive compounds	Health effect	Reference
Extra animal protein			
Grasshopper (<i>Sphenarium purpurascens</i>)	Bioactive peptides, linoleic acid (ω -6), α -linolenic acid (ω -3).	Antioxidant, hypocholesterolemic, intestinal health.	(91)
Egg	Lutein, zeaxanthin, monounsaturated, polyunsaturated fatty acids, carotenoids.	Hypocholesterolemic, antioxidant, cardioprotective.	(92)
Chicken	Bioactive peptides, oleic acid (ω -9).	Cardioprotective, glycemic control.	(93)
Fish	α -Linolenic acid (EPA, DHA).	Antioxidant, anti-inflammatory, neuroprotective, cardioprotective, antimicrobial, hepatoprotective.	(94)
Tubers			
Yucca (<i>Manihot esculenta</i>)	Resistant starch, polyphenols, saponins, flavonoids, tanins, terpenoids.	Antibacterial, antioxidant, glycemic control, anti-inflammatory, anticancer, anti-diarrheal, hypocholesterolemic.	(95)
Sweet potato (<i>Ipomoea batatas</i>)	Carotenoids, fiber, polyphenols, anthocyanins, linoleic acid (ω -6).	Antioxidant, glycemic control, cardioprotective, anti-inflammatory, anticancer, antimicrobial.	(96)
Sweeteners			
Honey (<i>Apis mellifera</i> L.)	Flavonoids, polyphenols, carotenoids, enzymes.	Antioxidant, anti-inflammatory, glycemic control, antimicrobial, antiparasitic, antiviral, anticancer.	(97)

Bioactive components are listed in descending order of content.

1 Mechanisms of action: the matrix effect

Specific food-processing interactions and enzymatic inhibitions drive the physiological impact of the Milpa. For instance, the nixtamalization of maize not only enhances calcium bioavailability but modifies the anthocyanin profile, increasing the concentration of cyanidin-3-glucoside (C3G), a flavonoid with proven insulin-sensitizing properties (28, 29). Concurrently, the consumption of common beans provides specific inhibitors of α -amylase and α -glucosidase. These bioactive peptides blunt postprandial glycemic excursions by delaying carbohydrate hydrolysis (30, 31). When combined with the lipid profile of pumpkin seeds (rich in PUFAs and phytosterols), this triad creates a metabolic buffer that optimizes glucose homeostasis and lipid metabolism.

2 Bioactive architecture

The health benefits are mechanistically attributable to a distinct phytochemical profile that exerts pleiotropic effects (23, 32). Key bioactive groups include:

- Phenolic compounds: specifically, flavonoids, tannins, and phenolic acids. These act as potent antioxidants and modulators of inflammatory pathways (e.g., NF- κ B inhibition) (23).
- Phytosterols: including β -sitosterol, campesterol, and stigmasterol. These compounds compete with cholesterol for

micellar absorption in the gut, effectively improving lipid profiles (23, 33).

- Carotenoids: such as zeaxanthin, lutein, and β -carotene. Beyond their role in ocular health, they function as signaling molecules in adipose tissue regulation (17).
- Bioactive peptides: derived from plant proteins (beans and amaranth), exhibiting antihypertensive (ACE-inhibitory) and antithrombotic activities (34).
- Dietary fiber and fatty acids: the synergy between soluble fiber and oleic/linoleic acids supports satiety signaling and improved insulin receptor sensitivity (35, 36).

3 Clinical evidence: from prevention to management

Accumulating evidence demonstrates that adherence to the Milpa diet—and the broader Traditional Mexican Diet (TMD)—confers protection against the spectrum of chronic non-communicable diseases (NCDs). Clinical and epidemiological studies have documented significant risk reductions in:

- Cardiometabolic conditions: including Type 2 Diabetes (T2D), dyslipidemias, Metabolic dysfunction-associated steatotic liver disease (MASLD), and obesity (13, 16, 37–39).
- Systemic pathologies: such as cardiovascular disease (CVD), chronic kidney disease (CKD), and specific cancers, as well as autoimmune conditions like celiac disease (5, 17, 40).

4 Precision nutrition and the gut-brain axis

Finally, the Milpa diet represents a frontier for Precision Nutrition. Its plant-forward composition serves as a high-quality substrate for the gut microbiota, promoting the production of Short-Chain Fatty Acids (SCFAs), such as butyrate. This mechanism not only reinforces gut barrier integrity but also modulates the gut-brain axis (41, 42). Crucially, these benefits are amplified by the synergy of functional foods inherent to the system. Beyond providing essential nutrients, the diversity of Milpa crops delivers a complex matrix of bioactive compounds with antioxidant and anti-inflammatory properties that play a key role in mental health (23, 43). As research advances in nutrigenomics, the Milpa stands as a scalable model to address genetic susceptibilities (e.g., *MTHFR* or *ABCA1* variants) through these culturally relevant dietary interventions (13, 44).

Clinical and epidemiological evidence: moving beyond the western diet

The Western Diet (WD)—characterized by excessive caloric density, nutrient deficiency, and a reliance on ultra-processed foods (UPFs)—acts as a primary driver of metabolic inflexibility and systemic chronic inflammation (2). The public health ramifications of this dietary transition are catastrophic, reflected in the escalating global mortality burden linked to poor nutritional quality (1, 5).

Current epidemiological estimates paint a stark picture of this “metabolic collision.” Approximately 6.58 million deaths from cardiovascular disease (CVD) are attributable to dietary risks, specifically the excessive intake of sodium, trans fats, and processed meats (6). Furthermore, nutritional imbalances—such as low whole-grain and fruit consumption—are directly linked to 381,000 deaths from Type 2 Diabetes (T2D) (1) and 317,000 deaths among patients with chronic kidney disease (CKD) (5). Moreover, the obesity epidemic, fueled by the WD, now accounts for nearly 357,000 cancer deaths annually (45, 46). Extending beyond somatic pathology, this dietary pattern drives neuronal alterations that have opened a critical debate regarding the addictive potential of ultra-processed foods, further underscoring the emerging intersection between nutrition and mental health (47).

Against this backdrop, the Milpa diet emerges not merely as a cultural alternative but as a potent physiological corrective. While clinical literature often categorizes these interventions as “Traditional Mexican Diet” (TMD) or “Genome-Based Diets,” they are fundamentally rooted in the Milpa triad and its associated agrobiodiversity. To systematically validate this, existing studies provide robust evidence across various methodological designs and sample populations:

Cross-sectional epidemiological evidence demonstrates that reintroducing these pre-Hispanic staples generates a favorable lipidomic profile among adult populations, significantly reducing total cholesterol, LDL-c, VLDL-c, and triglycerides—key drivers of atherogenic risk (37, 44). Furthermore, randomized crossover feeding trials in women of Mexican descent reveal that the diet exerts a regulatory effect on glucose homeostasis, reducing insulin levels and HOMA-IR (44). Significantly, these controlled

trials show it lowers serum concentrations of high-sensitivity C-reactive protein (hsCRP), thereby dampening the systemic low-grade inflammation characteristic of the WD (48).

Crucially, the Milpa diet exhibits potential as a tool for population-specific precision nutrition. Clinical intervention studies indicate that its metabolic benefits are particularly pronounced in individuals carrying risk polymorphisms common in Amerindian and Mestizo populations (e.g., *MTHFR* 677 CT, *ABCA1* R230C, and *APOE* ϵ 4). This suggests that the Milpa pattern may mitigate the deleterious phenotypic expression of these high-risk genotypes (44, 48).

Mechanistically, *in vivo* animal models utilizing rats fed a “Pre-Hispanic Diet” formulation confirm these observations at the molecular level. Consumption of this pattern is linked to the upregulation of key metabolic regulators, including PPAR- α (fatty acid oxidation), UCP-1 (thermogenesis), and the mitochondrial biogenesis coactivator PGC-1 α .

This molecular cascade results in reduced adipocyte hypertrophy and lower hepatic triglyceride accumulation (49). Concomitantly, the diet improves the cellular redox state by decreasing Reactive Oxygen Species (ROS) and oxidized proteins while optimizing the GSSG/GSH ratio. Notably, these neuroprotective and antioxidant mechanisms promote systemic conditions associated with cognitive health (43, 49, 50).

The Milpa diet for planetary health

The current anthropogenic crisis reveals a dangerous paradox: while approximately 82% of the human caloric intake comes from plants (51), global dietary diversity has collapsed. Despite the existence of over 6,000 edible plant species, 60% of agricultural production is effectively monopolized by just nine crops (e.g., maize, wheat, rice, soy). These are produced almost exclusively via high-input monocultures intended not for direct human nutrition but for the manufacture of ultra-processed food ingredients (3, 7). This industrial model drives soil depletion, biodiversity loss, and creates a critical rift between human consumption and environmental sustainability.

In stark contrast, the Milpa system functions as a model of ecological intensification, capable of sustaining productivity even under adverse climatic conditions (20). Unlike monocultures that act as biological vacuums, the Milpa thrives on the specific niche complementarity of the Mesoamerican triad:

- **Maize (C4 architecture):** acts as the vertical axis, optimizing solar radiation capture via C4 photosynthesis while providing physical support for climbing legumes.
- **Beans (nitrogen fixation):** establishes a symbiotic relationship with *Rhizobium* bacteria to fix atmospheric nitrogen, naturally refertilizing the soil and enriching the system for associated crops.
- **Squash (microclimate regulation):** its broad leaves function as a living mulch, intercepting sunlight to suppress competitive weeds, maintaining soil moisture, and secreting cucurbitacins—biochemical compounds that provide allelopathic protection against pests (19, 20, 52).

This biological synergy generates a robust microclimate that optimizes the “Land Equivalent Ratio” (LER). The diverse aerial architecture and complex root stratification enable efficient resource partitioning, significantly enhancing resilience against droughts and pest outbreaks compared with simplified systems (18, 19).

Consequently, the Milpa stands as an agroecological system that embodies the principles of Planetary Health, effectively linking sustainable food production with ecosystem conservation and human wellbeing (15, 53). Its historical persistence serves as a rigorous proof of concept: it demonstrates that it is possible to harmonize nutritional needs with planetary boundaries, offering a sustainability model rooted in traditional knowledge and respectful coexistence with nature.

Beyond its response to routine anthropogenic impacts, the systemic stability of the Milpa positions it as a critical nutritional solution during climate-induced emergencies and natural disasters. As global, highly centralized supply chains become increasingly vulnerable to severe climate shocks, the reliance on imported ultra-processed foods poses a severe threat to regional food security. Developing and safeguarding culture-specific dietary profiles like the Milpa serves as a vital contingency strategy. By maintaining local agricultural knowledge and utilizing indigenous, drought-resistant crops, communities establish a reliable, nutrient-dense safety net when industrial food systems fail. In this light, the preservation of local culinary heritage transcends cultural identity; it becomes a tangible, life-saving strategy for food security and disaster resilience in an era of climate unpredictability.

Discussion: challenges, opportunities, and the strategic revaluation of the Milpa

While the physiological and ecological evidence supporting the Milpa diet is compelling, its effective reimplementation faces systemic barriers rooted in the “industrial lock-in” of current food systems. The globalization of agriculture has homogenized dietary landscapes, creating environments where ultra-processed foods are ubiquitous and often cheaper than fresh staples (7, 8). However, this transition has created a critical dissonance: while globalization offers dietary diversity, nutrition must not be governed by transient trends. The definition of a “correct diet” remains immutable in its biological principles—complete, varied, sufficient, safe, and adequate—but in the Anthropocene, it must imperatively include two new dimensions: sustainability and cultural pertinence.

In this context, the Milpa diet represents the closest approximation to population-based precision nutrition. While global frameworks like the EAT-Lancet planetary health diet provide vital universal targets for sustainable eating, and the Mediterranean diet serves as a recognized gold standard for cardiometabolic health, the Milpa diet offers a critical localized counterpart. Universal models, while highly valuable, often lack the cultural resonance and specific genomic synchronization required for diverse populations. For instance, while the Mediterranean diet achieves cardiovascular protection largely through olive oil and

wine, the Milpa achieves comparable functional synergy through the lipid profiles of pumpkin seeds, avocados, and the bioactive peptides of diverse legumes.

For Mesoamerican and Mestizo populations, adopting the Milpa pattern is not merely a cultural choice but a biological imperative; it respects the genetic architecture of native peoples, offering a metabolic substrate to which they are historically and evolutionarily adapted (44, 54). Unlike imported dietary models that may exacerbate metabolic discordance, the Milpa offers a “genomic synchronization” that valorizes ancestral wisdom not just as folklore, but as empirically valid preventive medicine.

It is important to note the contextual boundaries of this system. The exact crop matrix of the Milpa is highly scalable and therapeutically optimal within Mesoamerica and for Hispanic/Latine populations globally. However, its direct agricultural implementation may not be ecologically applicable or culturally resonant in fundamentally different edaphoclimatic zones (e.g., Northern Europe or Arid Asia). In such contexts, the Milpa serves not as a literal dietary export, but as a conceptual paradigm. It demonstrates how reclaiming any region’s native, agroecological polycultures (e.g., the Nordic diet or traditional Asian intercropping systems) can simultaneously resolve local health and environmental crises.

Adopting this pattern generates a virtuous cycle of nutritional and agroecological synergy. By consuming local polycultures, we amplify functionality: the diet becomes more bioactive, while the land becomes more resilient. This approach transcends the reductionist view of nutrition as mere “food intake,” repositioning it as a multidimensional phenomenon encompassing culture, society, the economy, health, and prevention. It fosters local economies, strengthens food sovereignty, and dignifies the rural sector, countering the stigmatization that has marginalized indigenous foodways (14, 18, 55).

To operationalize this salutogenic vision—focused on generating health rather than managing disease—scientific knowledge must transcend the laboratory. Effective science communication serves as the critical bridge to translate these findings into robust public policies. We propose a systemic intervention framework:

1. Agroecological policy reorientation: shift subsidies toward polycultural, small-scale farming to conserve native germplasm *in-situ* and ensure the resilience of the diet’s genetic base (51).
2. From superfoods to biocultural heritage: educational strategies must dismantle the “superfood” narrative (which leads to commodification and elitism) and replace it with a “Biocultural Heritage” model. This validates indigenous food systems as scientifically superior strategies for community health (15).
3. Sustainable public procurement: incorporating Milpa foods into school feeding programs and public institutions is a potent tool for normalizing these nutrient-dense foods, ensuring that the “correct diet” is accessible to the most vulnerable.
4. Mindful eating as a promotion strategy: since “real food” lacks the massive advertising machinery of ultra-processed products, we must actively cultivate Mindful Eating not merely as a clinical tool, but as a holistic public health necessity. This approach reawakens the conscious connection between the eater, the origin of the food, and the act of nourishment (56).

By emphasizing the sensory and cultural richness of the Milpa, we can counter the industrial marketing narrative and empower individuals to intentionally choose health and sustainability over convenience.

Ultimately, the revitalization of the Milpa diet demands treating individual health, community well-being, and planetary sustainability as indivisible outcomes.

Conclusion: a path toward sustainable health and resilience

The Milpa diet transcends the traditional meal concept to become a sophisticated, scalable strategy for planetary health. By synchronizing human metabolic needs—specifically adhering to the genetic adaptations of Mesoamerican populations—with agroecological resilience, it effectively resolves the dichotomy between nutrition and sustainability. Revitalizing this system is not an exercise in nostalgia, but a critical innovation for the Anthropocene. Ultimately, the Milpa offers a proven, salutogenic blueprint to navigate the global syndemic, demonstrating that the most resilient future for nutrition lies in the scientific revalorization of our ancestral roots.

Author contributions

RF-D: Project administration, Writing – review & editing, Validation, Supervision, Methodology, Writing – original draft, Data curation, Investigation, Conceptualization. MS-J: Data curation, Methodology, Visualization, Investigation, Writing – review & editing, Conceptualization, Project administration. CH-C: Conceptualization, Writing – review & editing, Supervision, Investigation. IV-M: Writing – review & editing, Methodology, Visualization, Investigation, Validation, Project administration. GR-S: Conceptualization, Data curation, Methodology, Writing – review & editing, Supervision, Investigation. RR-R: Conceptualization, Writing – review & editing, Investigation, Supervision, Validation, Methodology, Visualization. GP-H: Writing – review & editing, Conceptualization, Visualization, Investigation. MÁ-R: Validation, Writing – review & editing, Conceptualization, Investigation.

References

1. Ma H, Wang M, Qin C, Shi Y, Mandizadza OO, Ni H, et al. Trends in the burden of chronic diseases attributable to diet-related risk factors from 1990 to 2021 and the global projections through 2030: a population-based study. *Front Nutr.* (2025) 12:1570321. doi: 10.3389/fnut.2025.1570321
2. Clemente-Suárez VJ, Beltrán-Velasco AI, Redondo-Flórez L, Martín-Rodríguez A, Tornero-Aguilera JF. Global impacts of western diet and its effects on metabolism and health: a narrative review. *Nutrients.* (2023) 15:2749. doi: 10.3390/nu15122749
3. Vega-Mejía N, Ponce-Reyes R, Martínez Y, Carrasco O, Cerritos R. Implications of the western diet for agricultural production, health and climate Change. *Front Sustain Food Syst.* (2018) 2:88. doi: 10.3389/fsufs.2018.00088
4. GBD 2021 Adolescent BMI Collaborators. Global, regional, and national prevalence of child and adolescent overweight and obesity, 1990–2021, with forecasts to 2050: a forecasting study for the Global Burden of Disease Study 2021. *Lancet.* (2025) 405:785–812. doi: 10.1016/S0140-6736(25)00397-6

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Acknowledgments

Author MAS-J (CVU: 1040323) thanks the Secretaría de Ciencia, Humanidades, Tecnología e Innovación (SECIHTI) (SECIHTI) for the doctoral scholarship. Additionally, authors RF-D (CVU: 714861), IV-M (CVU: 863439), GUR-S (CVU: 714866), and RR-R (CVU: 785735) acknowledge the support received through the Postdoctoral Stays in Mexico program.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

5. Wei N, Yang M, Zheng P, Xu J. Burden and inequalities of chronic kidney disease attributable to diet globally, regionally and temporally, 1990–2021. *Front Nutr.* (2025) 12:1592389. doi: 10.3389/fnut.2025.1592389
6. Vaduganathan M, Mensah GA, Turco JV, Fuster V, Roth GA. The global burden of cardiovascular diseases and risk: a compass for future health. *J Am Coll Cardiol.* (2022) 80:2361–71. doi: 10.1016/j.jacc.2022.11.005
7. Leite FHM, Khandpur N, Andrade GC, Anastasiou K, Baker P, Lawrence M, et al. Ultra-processed foods should be central to global food systems dialogue and action on biodiversity. *BMJ Glob Health.* (2022) 7:e008269. doi: 10.1136/bmjgh-2021-008269
8. Fardet A, Rock E. Ultra-processed foods and food system sustainability: what are the links? *Sustainability.* (2020) 12:6280. doi: 10.3390/su12156280
9. United Nations Convention to Combat Desertification (UNCCD). *Global Land Outlook 2: Land Restoration for Recovery and Resilience*. Bonn: UNCCD. (2022). Available online at: https://www.unccd.int/sites/default/files/2022-04/UNCCD_GLO2_low-res_2.pdf (Accessed January 2, 2026).
10. Fanzo J, Rudie C, Sigman I, Grinspoon S, Benton TG, Brown ME, et al. Sustainable food systems and nutrition in the 21st century: a report from the 22nd annual harvard nutrition obesity symposium. *Am J Clin Nutr.* (2022) 115:18–33. doi: 10.1093/ajcn/nqab315
11. Aydogdu GS, Karadag MG. The two dimensions of nutrition for the planet: environment and health. *Curr Nutr Rep.* (2025) 14:49. doi: 10.1007/s13668-025-00642-3
12. Sparling TM, Offner C, Deeney M, Denton P, Bash K, Juel R, et al. Intersections of climate change with food systems, nutrition, and health: an overview and evidence map. *Adv Nutr.* (2024) 15:100274. doi: 10.1016/j.advnut.2024.100274
13. Huerta-Álvarez A, Arellano M, Chávez-Méndez CA, Carpinteyro-Espin P, Palacios-Reyes C, Pérez-Escobar J. Milpa diet for MASLD in mesoamerican populations: feasibility, advantages, and future perspectives. *Life.* (2025) 15:812. doi: 10.3390/life15050812
14. Mota-Cruz C, Casas A, Ortega-Paczka R, Perales H, Vega-Peña E, Bye R. Milpa, a long-standing polyculture for sustainable agriculture. *Agriculture.* (2025) 15:1737. doi: 10.3390/agriculture15161737
15. Benrey B, Bustos-Segura C, Grof-Tisza P. The mesoamerican milpa system: Traditional practices, sustainability, biodiversity, and pest control. *Biol Control.* (2024) 198:105637. doi: 10.1016/j.biocontrol.2024.105637
16. Birute A, Leal-Escobar G, Espinosa-Cuevas Á, Mojica L, Kistler BM. Dieta de la Milpa: a culturally-concordant plant-based dietary pattern for hispanic/latine people with chronic kidney disease. *Nutrients.* (2024) 16:574. doi: 10.3390/nu16050574
17. Sánchez-Velázquez OA, Luna-Vital DA, Morales-Hernández N, Contreras J, Villaseñor-Tapia EC, Fragosso-Medina JA, et al. Nutritional, bioactive components and health properties of the Milpa triad system seeds (corn, common bean and pumpkin). *Front Nutr.* (2023) 10:1169675. doi: 10.3389/fnut.2023.1169675
18. Vazeux-Blumental N, Manicacci D, Tenaillon M. The milpa, from Mesoamerica to present days, a multicropping traditional agricultural system serving agroecology. *C R Biol.* (2024) 347:159–73. doi: 10.5802/crbio.164
19. Zizumbo-Villarreal D, Flores-Silva A, Colunga-García Marín P. The Archaic diet in Mesoamerica: incentive for Milpa development and species domestication. *Econ Bot.* (2012) 66:328–43. doi: 10.1007/s12231-012-9212-5
20. Fonteyne S, Castillo Caamal JB, Lopez-Ridaura S, Van Loon J, Balbuena JE, Osorio Alcalá L, et al. Review of agronomic research on the Milpa, the traditional polyculture system of Mesoamerica. *Front Agron.* (2023) 5:1115490. doi: 10.3389/fagro.2023.1115490
21. Rojas-Sánchez B, de los Santos-Villalobos S, Valdez Alarcón JJ, Chávez-Bárceñas AT, Orozco-Mosqueda MC, Santoyo G. Optimizing milpa agrosystems with beneficial microbes and their ecological interactions: a review. *Discover Appl Sci.* (2025) 7:104. doi: 10.1007/s42452-025-06503-6
22. Odukoya JO, De Saeger S, De Boevre M, Adegoke GO, Audenaert K, Croubels S, et al. Influence of nixtamalization cooking ingredients on the minerals composition of nixtamalized maize and sorghum. *J Cereal Sci.* (2022) 103:103373. doi: 10.1016/j.jcs.2021.103373
23. Méndez-Flores OG, Ochoa-Díaz López H, Castro-Quezada I, Olivo-Vidal ZE, García-Miranda R, Rodríguez-Robles U, et al. The Milpa as a supplier of bioactive compounds: a review. *Food Rev Int.* (2023) 39:1359–76. doi: 10.1080/87559129.2021.1934001
24. Celmeli T, Sari H, Canci H, Sari D, Adak A, Eker T, et al. The nutritional content of common bean (*Phaseolus vulgaris* L.) landraces in comparison to modern varieties. *Agronomy.* (2018) 8:166. doi: 10.3390/agronomy8090166
25. Huerta-Reyes M, Tavera-Hernández R, Alvarado-Sansininea JJ, Jiménez-Estrada M. Selected species of the Cucurbitaceae family used in Mexico for the treatment of diabetes mellitus. *Molecules.* (2022) 27:3440. doi: 10.3390/molecules2713440
26. Adnan M, Gul S, Batool S, Fatima B, Rehman A, Yaqoob S, et al. A review on the ethnobotany, phytochemistry, pharmacology and nutritional composition of *Cucurbita pepo* L. *J Phytopharmacol.* (2017) 6:133–9. doi: 10.31254/phyto.2017.6211
27. Hernandez-Perez T, Gómez-García MR, Valverde ME, Paredes-López O. *Capsicum annum* (hot pepper): an ancient Latin-American crop with outstanding bioactive compounds and nutraceutical potential. *Compr Rev Food Sci Food Saf.* (2020) 19:2972–93. doi: 10.1111/1541-4337.12634
28. Mora-Rochín S, Gaxiola-Cuevas N, Gutiérrez-Urbe JA, Milán-Carrillo J, Milán-Noris EM, Reyes-Moreno C, et al. Effect of traditional nixtamalization on anthocyanin content and profile in Mexican blue maize (*Zea mays* L.) landraces. *LWT-Food Sci Technol.* (2016) 68:563–9. doi: 10.1016/j.lwt.2016.01.009
29. Martín MÁ, Ramos S. Dietary flavonoids and insulin signaling in diabetes and obesity. *Cells.* (2021) 10:1474. doi: 10.3390/cells10061474
30. Oseguera-Toledo ME, de Mejía EG, Amaya-Llano SL. Hard-to-cook bean (*Phaseolus vulgaris* L.) proteins hydrolyzed by alcalase and bromelain produced bioactive peptide fractions that inhibit targets of type-2 diabetes and oxidative stress. *Food Res Int.* (2015) 76:839–51. doi: 10.1016/j.foodres.2015.07.046
31. Mojica L, Chen K, de Mejía EG. Impact of commercial precooking of common bean (*Phaseolus vulgaris*) on the generation of peptides, after pepsin–pancreatin hydrolysis, capable to inhibit dipeptidyl peptidase-IV. *J Food Sci.* (2015) 80:H188–98. doi: 10.1111/1750-3841.12726
32. Almaguer González JA, García Ramírez HJ, Vargas Vite V, Padilla Mirazo M. *Fortalecimiento de la Salud Con Comida, Ejercicios y Buen Humor: La Dieta de la Milpa Modelo de Alimentación Mesoamericana Saludable y Culturalmente Pertinente*. Ciudad de México: Secretaría de Salud (2018). Available at: <https://pesquisa.bvsalud.org/portal/resource/pt/biblio-880586>
33. Ríos-Hoyo A, Romo-Araiza A, Meneses-Mayo M, Gutiérrez-Salmeán G. Prehispanic functional foods and nutraceuticals in the treatment of dyslipidemia associated to cardiovascular disease: a mini-review. *Int J Vitam Nutr Res.* (2017) 87:85–98. doi: 10.1024/0300-9831/a000290
34. Hernández-Pérez T, Paredes-López O. Ancient Latin-American food crops: an overview of their nutraceutical and antiobesity peptides. *Food Sci Tech Int.* 2025;0(0). doi: 10.1177/10820132251319934
35. López-Romero P, Pichardo-Ontiveros E, Avila-Nava A, Vázquez-Manjarrez N, Tovar AR, Pedraza-Chaverri J, et al. The effect of nopal (*Opuntia ficus indica*) on postprandial blood glucose, incretins, and antioxidant activity in Mexican patients with type 2 diabetes after consumption of two different composition breakfasts. *J Acad Nutr Diet.* (2014) 114:1811–8. doi: 10.1016/j.jand.2014.06.352
36. Alatorre-Cruz JM, Carreño-López R, Alatorre-Cruz GC, Paredes-Esquivel LJ, Santiago-Saenz YO, Nieva-Vázquez A. Traditional Mexican food: phenolic content and public health relationship. *Foods.* (2023) 12:1233. doi: 10.3390/foods12061233
37. Valerino-Perea S, Armstrong MEG, Papadaki A. Adherence to a traditional Mexican diet and non-communicable disease-related outcomes: secondary data analysis of the cross-sectional Mexican national health and nutrition survey. *Br J Nutr.* (2023) 129:1266–79. doi: 10.1017/S0007114522002331
38. Santiago-Torres M, Kratz M, Lampe JW, Tapsoba JD, Breymeyer KL, Levy L, et al. Metabolic responses to a traditional Mexican diet compared with a commonly consumed US diet in women of Mexican descent: a randomized crossover feeding trial. *Am J Clin Nutr.* (2016) 103:366–74. doi: 10.3945/ajcn.115.119016
39. Loroña NC, Santiago-Torres M, Lopez-Pentecost M, et al. Traditional Mexican dietary pattern and cancer risk among women of Mexican descent. *Cancer Causes Control.* (2024) 35:887–96. doi: 10.1007/s10552-024-01849-5
40. Lopez-Pentecost M, Tamez M, Mattei J, Jacobs ET, Thomson CA, Garcia DO. Adherence to a traditional Mexican diet is associated with lower hepatic steatosis in US-born hispanics of Mexican descent with overweight or obesity. *Nutrients.* (2023) 15:4997. doi: 10.3390/nu15234997
41. Guevara-Cruz M, Tovar AR, Aguilar-Salinas CA, Medina-Vera I, Gil-Zenteno I, Hernández-Viveros I, et al. A dietary pattern including nopal, chia seed, soy protein, and oat reduces serum triglycerides and glucose intolerance in patients with metabolic syndrome. *J Nutr.* (2012) 142:64–9. doi: 10.3945/jn.111.147447
42. Guizar-Heredia R, Noriega LG, Rivera AL, Resendis-Antonio O, Guevara-Cruz M, Torres N, et al. A new approach to personalized nutrition: postprandial glycemic response and its relationship to gut microbiota. *Arch Med Res.* (2023) 54:176–88. doi: 10.1016/j.arcmed.2023.02.007
43. Fernández-Demeneghi R, Domínguez-Pantoja M, Martínez-Moreno AG, Vargas-Moreno I, Ramírez-Rodríguez R. Positioning berries in nutritional psychiatry: potential for prevention and co-therapy in mental health. *Front Behav Neurosci.* (2025) 19:1622242. doi: 10.3389/fnbeh.2025.1622242
44. Ojeda-Granados C, Panduro A, Rivera-Iníguiz I, Sepúlveda-Villegas M, Roman S. A regionalized genome-based Mexican diet improves anthropometric and metabolic parameters in subjects at risk for obesity-related chronic diseases. *Nutrients.* (2020) 12:645. doi: 10.3390/nu12030645
45. Figlioli G, Piovani D, Tsantes AG, Pugliese N, Nikolopoulos GK, Hassan C, et al. Burden of cancer attributable to high body mass index: a systematic analysis of the global burden of disease study 2021. *Clin Nutr.* (2025) 48:144–52. doi: 10.1016/j.clnu.2025.04.002
46. Li X, Li H. Global, national, and regional burden of cancer attributable to dietary risk: results from the global burden of disease study 2021. *BMC Public Health.* (2025) 25:3244. doi: 10.1186/s12889-025-24570-7

47. Gearhardt AN, DiFeliceantonio AG. Highly processed foods can be considered addictive substances based on established scientific criteria. *Addiction*. (2023) 118:589–98. doi: 10.1111/add.16065
48. Santiago-Torres M, Tinker LF, Allison MA, Breymeyer KL, Garcia L, Kroenke CH, et al. Development and use of a Traditional Mexican Diet Score in relation to systemic inflammation and insulin resistance among women of Mexican descent. *J Nutr*. (2015) 145:2732–40. doi: 10.3945/jn.115.21353859
49. Avila-Nava A, Noriega LG, Tovar AR, Granados O, Perez-Cruz C, Pedraza-Chaverri J, et al. Food combination based on a pre-hispanic Mexican diet decreases metabolic and cognitive abnormalities and gut microbiota dysbiosis caused by a sucrose-enriched high-fat diet in rats. *Mol Nutr Food Res*. 2017;61(1). doi: 10.1002/mnfr.201501023
50. Sánchez-Tapia M, Aguilar-López M, Pérez-Cruz C, Pichardo-Ontiveros E, Wang M, Donovan SM, et al. Nopal (*Opuntia ficus indica*) protects from metabolic endotoxemia by modifying gut microbiota in obese rats fed high fat/sucrose diet. *Sci Rep*. (2017) 7:4716. doi: 10.1038/s41598-017-05096-4
51. FAO. *The State of the World's Biodiversity for Food and Agriculture*. Rome: FAO. (2019). Available online at: <https://openknowledge.fao.org/server/api/core/bitstreams/50b79369-9249-4486-ac07-9098d07df60a/content> (Accessed January 2, 2026).
52. Grof-Tisza P, Muller MH, González-Salas R, Bustos-Segura C, Benrey B. The Mesoamerican Milpa agroecosystem fosters greater arthropod diversity compared to monocultures. *Agric Ecosyst Environ*. (2024) 372:109074. doi: 10.1016/j.agee.2024.109074
53. González-Esquivel CE, Briones-Guzmán C, Tovar-López E, et al. Sustainability evaluation of contrasting Milpa systems in the Yucatán Peninsula, Mexico. *Environ Dev Sustain*. (2025) 27:9233–55. doi: 10.1007/s10668-023-04281-y
54. De Cosmi V, Mazzocchi A, Milani GP, Agostoni C. Dietary patterns vs. dietary recommendations. *Front Nutr*. (2022) 9:883806. doi: 10.3389/fnut.2022.883806
55. Rodríguez-Robayo KJ, Méndez-López ME, Molina-Villegas A, Juárez L. What do we talk about when we talk about Milpa? A conceptual approach to the significance, topics of research and impact of the mayan Milpa system. *J Rural Stud*. (2020) 77:47–54. doi: 10.1016/j.jrurstud.2020.04.029
56. Fernández-Demeneghi R, Sánchez-Bizama J, Martínez-Moreno AG, Vargas Moreno I, Ramírez-Rodríguez R, Puga Olguín A, et al. Mindful eating as the next therapeutic frontier in nutritional psychiatry. *Front Nutr*. (2026) 13:1726847. [Preprint] doi: 10.3389/fnut.2026.1726847
57. Kasali FM, Tusiimire J, Kadima JN, Agaba AG. Ethnomedical uses, chemical constituents, and evidence-based pharmacological properties of *Chenopodium ambrosioides* L.: extensive overview. *Future J Pharm Sci*. (2021) 7:153. doi: 10.1186/s43094-021-00306-3
58. Segura-Campos MR, Ruiz-Ruiz JC, Chel-Guerrero LA, Betancur-Ancona DA. *Capsicum chinense*: composition and functional properties. In: Kristberg Kristbergsson, Semih Ötles, editors. *Functional Properties of Traditional Foods. Integrating Food Science and Engineering Knowledge Into the Food Chain* Vol 12. New York: Springer. (2016). p. 289–92. doi: 10.1007/978-1-4899-7662-8_20
59. Santiago-López L, García HS, González-Córdova AF, Vallejo-Córdova B, Hernández-Mendoza A. Antioxidant and anti-inflammatory properties of huauzontle (*Chenopodium berlandieri* subsp. *nuttalliae*, Chenopodiaceae) fermented by *Lactiplantibacillus plantarum* L p22. *Acta Bot Mex*. (2023) 130:e2161. doi: 10.21829/abm130.2023.216160
60. El-Mostafa K, El Kharrassi Y, Badreddine A, Andreoletti P, Vamecq J, El Kebaj MHS, et al. Nopal cactus (*Opuntia ficus-indica*) as a source of bioactive compounds for nutrition, health and disease. *Molecules*. (2014) 19:14879–901. doi: 10.3390/molecules19091487968
61. Valdez-Morales M, Céspedes-Carlos L, Valverde ME, Ramírez-Chávez E, Paredes-López O. Phenolic compounds, antioxidant activity and lipid profile of huitlacoche mushroom (*Ustilago maydis*) produced in several maize genotypes at different stages of development. *Plant Foods Hum Nutr*. (2016) 71:436–43. doi: 10.1007/s11130-016-0572-3
62. Ali MY, Sina AAI, Khandker SS, Neesa L, Tanvir EM, Kabir A, et al. Nutritional composition and bioactive compounds in tomatoes and their impact on human health and disease: a review. *Foods*. (2021) 10:45. doi: 10.3390/foods10010045
63. Shenstone E, Lippman Z, Van Eck J. A review of nutritional properties and health benefits of *Physalis* species. *Plant Foods Hum Nutr*. (2020) 75:316–25. doi: 10.1007/s11130-020-00821-3
64. Di Lorenzo R, Castaldo L, Sessa R, Ricci L, Vardaro E, Izzo L, et al. Chemical profile and promising applications of *Cucurbita pepo* L. flowers. *Antioxidants (Basel)*. (2024) 13:1476. doi: 10.3390/antiox13121476
65. Vieira EF, Pinho O, Ferreira IMPLVO, Delerue-Matos C. Chayote (*Sechium edule*): a review of nutritional composition, bioactivities and potential applications. *Food Chem*. (2019) 275:557–68. doi: 10.1016/j.foodchem.2018.09.146
66. García-Vásquez R, Vera-Guzmán AM, Carrillo-Rodríguez JC, Pérez-Ochoa ML, Aquino-Bolaños EN, Alba-Jiménez JE, et al. Bioactive and nutritional compounds in fruits of pepper (*Capsicum annum* L.) landraces conserved among indigenous communities from Mexico. *AIMS Agric Food*. (2023) 8:832–50. doi: 10.3934/agrfood.2023044
67. Mateos-Maces L, Chávez-Servia JL, Vera-Guzmán AM, et al. Edible leafy plants from Mexico as sources of antioxidant compounds, and their nutritional, nutraceutical and antimicrobial potential: a review. *Antioxidants (Basel)*. (2020) 9:531. doi: 10.3390/antiox9060541
68. Ghorani V, Saadat S, Khazdair MR, Gholamnezhad Z, El-Seedi H, Boskabady MH. Phytochemical characteristics and anti-inflammatory, immunoregulatory, and antioxidant effects of *Portulaca oleracea* L.: a comprehensive review. *Evid Based Complement Alternat Med*. (2023) 2023:2075444. doi: 10.1155/2023/207544475
69. Mannino G, Serio G, Berteau CM, Chiarelli R, Lauria A, Gentile C. Phytochemical profile and antioxidant properties of the edible and non-edible portions of black sapote (*Diospyros digyna* Jacq.). *Food Chem*. (2022) 380:132137. doi: 10.1016/j.foodchem.2022.132137
70. Sánchez-Recillas E, Campos-Vega R, Pérez-Ramírez IF, Luzardo-Ocampo I, Cuéllar-Núñez ML, Vergara-Castañeda HA. Garambullo (*Myrtillocactus geometrizans*): effect of in vitro gastrointestinal digestion on the bioaccessibility and antioxidant capacity of phytochemicals. *Food Funct*. (2022) 13:4699–713. doi: 10.1039/D1FO04392G
71. de Oliveira AC, Moreira Mar J, Frota Corrêa R, Sanches EA, Campelo PH, Ramos AS, et al. *Pouteria* spp. fruits: health benefits of bioactive compounds and their potential for the food industry. *Food Res Int*. (2023) 164:113310. doi: 10.1016/j.foodres.2023.113310
72. Barreca D, Laganà G, Ficarra S, Tellone E, Leuzzi U, Galtieri A, et al. Evaluation of the antioxidant and cytoprotective properties of the exotic fruit *Annona cherimola* Mill. (*Annonaceae*). *Food Res Int*. (2011) 44:2302–10. doi: 10.1016/j.foodres.2011.02.031
73. Nishikito DF, Borges ACA, Laurindo LF, Otoboni AMMB, Direito R, Goulart RA, et al. Anti-inflammatory, antioxidant, and other health effects of dragon fruit and potential delivery systems for its bioactive compounds. *Pharmaceutics*. (2023) 15:159. doi: 10.3390/pharmaceutics15010159
74. Yahia EM, Gutiérrez-Orozco F, Arvizu-de León C. Phytochemical and antioxidant characterization of mamey (*Pouteria sapota* Jacq HE Moore & Stearn) fruit. *Food Res Int*. (2011) 44:2175–81. doi: 10.1016/j.foodres.2010.11.029
75. Nazhand A, Lucarini M, Durazzo A, Zaccardelli M, Cristarella S, Souto SB, et al. Hawthorn (*Crataegus* spp.): An updated overview on its beneficial properties. *Forests*. (2020) 11:564. doi: 10.3390/f1105056476
76. Kong YR, Jong YX, Balakrishnan M, Bok ZK, Weng JKK, Tay KC, et al. Beneficial role of Carica papaya extracts and phytochemicals on oxidative stress and related diseases: a mini review. *Biology*. (2021) 10:287. doi: 10.3390/biology100402877
77. Ramírez-Rodríguez Y, Ramírez V, Robledo-Márquez K, García-Rojas N, Rojas-Morales P, Arango N, et al. Stenocereus huastecorum-fruit juice concentrate protects against cisplatin-induced nephrotoxicity by nitric oxide pathway activity and antioxidant and antiapoptotic effects. *Food Res Int*. (2022) 160:111337. doi: 10.1016/j.foodres.2022.11133778
78. Sameh S, Al-Sayed E, Labib RM, Singab AN. Genus spondias: a phytochemical and pharmacological review. *Evid Based Complement Alternat Med*. (2018) 2018:5382904. doi: 10.1155/2018/538290483
79. Bangar S, Sharma N, Kaur H, Kaur M, Sandhu KS, Maqsood S, et al. A review of Sapidilla (*Manilkara zapota*) in human nutrition, health, and industrial applications. *Trends Food Sci Technol*. (2022) 127:319–34. doi: 10.1016/j.tifs.2022.05.016
80. Morales P, Ramírez-Moreno E, Sánchez-Mata MC, Carvalho AM, Ferreira ICR. Nutritional and antioxidant properties of pulp and seeds of two *Xoconostle cultivars* (*Opuntia joconostle* FAC Weber ex Diguët and *Opuntia matudae* Scheinvar) of high consumption in Mexico. *Food Res Int*. (2012) 46:279–85. doi: 10.1016/j.foodres.2011.12.031
81. Coria-Téllez AV, Montalvo-González E, Yahia EM, Obledo-Vázquez EN. *Annona muricata*: A comprehensive review on its traditional medicinal uses, phytochemicals, pharmacological activities, mechanisms of action and toxicity. *Arab J Chem*. (2018) 11:662–91. doi: 10.1016/j.arabjc.2016.01.004
82. Cabral B, Siqueira EMS, Bitencourt MAO, Lima MCJS, Lima AK, Ortmann CF, et al. Phytochemical study and anti-inflammatory and antioxidant potential of *Spondias mombin* leaves. *Rev Bras Farmacogn*. (2016) 26:304–11. doi: 10.1016/j.bjp.2016.02.002
83. Singhania N, Kumar R, Pramila, Bishnoi S, Ray AB, Diwan A. Bioactive properties and health benefits of amaranthus. In: *Harvesting Food from Weeds*. (2023). p. 351–83. doi: 10.1002/9781119793007.ch1084
84. Grancieri M, Martino HSD, Gonzalez de Mejia E. Chia seed (*Salvia hispanica* L.) as a source of proteins and bioactive peptides with health benefits: a review. *Compr Rev Food Sci Food Saf*. (2019) 18:480–99. doi: 10.1111/1541-4337.1242385
85. Adeleke BS, Babalola OO. Oilseed crop sunflower (*Helianthus annuus*) as a source of food: Nutritional and health benefits. *Food Sci Nutr*. (2020) 8:4666–84. doi: 10.1002/fsn3.178386
86. Ishaq S, Jafri L. Biomedical importance of cocoa (*Theobroma cacao*): significance and potential for the maintenance of human health. *Matrix Sci Pharma*. (2017) 1:1–5. doi: 10.26480/msp.01.2017.01.0587
87. Çiftçi S, Suna GÜ. Functional components of peanuts (*Arachis hypogaea* L.) and health benefits: a review. *Future Foods*. (2022) 5:100140. doi: 10.1016/j.fufo.2022.10014097

88. Turco I, Ferretti G, Bacchetti T. Review of the health benefits of Faba bean (*Vicia faba* L.) polyphenols. *J Food Nutr Res.* (2016) 55:283–93.
89. Aslani Z, Alipour B, Mirmiran P, Bahadoran Z. Lentil's (*Lens culinaris* L.) functional properties in prevention and treatment of non-communicable chronic diseases: a review *Int J Nutr Food Sci.* (2014) 4:15–20. doi: 10.11648/j.ijnfs.s.2015040201.14
90. Fonseca Duarte P, Alves Chaves M, Dellinghausen Borges C, Barboza Mendonça CR. Avocado: characteristics, health benefits and uses. *Ciênc Rural.* (2016) 46:747–54. doi: 10.1590/0103-8478cr20141516
91. Fernández-Sánchez F, García-Barradas O, Mendoza-López MR, Juárez-Trujillo N, Luna-Solano G, Romero-Luna HE, et al. Nutritional potential of dehydrated *Sphenarium rugosum* powder: an insight into fatty acids and minerals. *Food Chem.* (2025) 485:144489. doi: 10.1016/j.foodchem.2025.144489
92. Kishimoto Y, Sugihara N. Egg consumption and human health: a comprehensive review of the effects on serum lipids, antioxidant status, and cardiovascular outcomes. *J Poult Sci.* (2026) 63:2026001. doi: 10.2141/jpsa.2026001
93. Connolly G, Clark CM, Campbell RE, Byers AW, Reed JB, Campbell WW. Poultry consumption and human health: how much is really known? a systematically searched scoping review and research perspective. *Adv Nutr.* (2022) 13:2115–24. doi: 10.1093/advances/nmac074
94. Chen J, Jayachandran M, Bai W, Xu B. A critical review on the health benefits of fish consumption and its bioactive constituents. *Food Chem.* (2022) 369:130874. doi: 10.1016/j.foodchem.2021.130874
95. Mohidin S, Moshawih S, Hermansyah A, Asmuni MI, Shafqat N, Ming LC. Cassava (*Manihot esculenta* Crantz): a systematic review for the pharmacological activities, traditional uses, nutritional values, and phytochemistry. *J Evid-Based Integr Med.* (2023) 28:2515690X231206227. doi: 10.1177/2515690X231206227
96. Alam MK. A comprehensive review of sweet potato (*Ipomoea batatas* [L.] Lam): Revisiting the associated health benefits Trends. *Food Sci Technol.* (2021) 115:512–29. doi: 10.1016/j.tifs.2021.07.001
97. Ciaciosci D, Forbes-Hernández TY, Afrin S, Gasparrini M, Reboredo-Rodríguez P, Manna PP, et al. Phenolic compounds in honey and their associated health benefits: a review. *Molecules.* (2018) 23:2322. doi: 10.3390/molecules23092322